

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12 ASSESSMENT TASK 3

JUNE 2003

MATHEMATICS

Time Allowed: 70 minutes

Instructions:

- * Attempt all questions.
- * Answers to be written on the paper provided.
- * Start each question on a new page.
- * All necessary working should be shown.
- * Marks may not be awarded for careless or badly arranged working.
- * This question paper must be stapled on top of your answers.
- * Marks shown are for guidance and may be changed slightly if needed.
- * Standard integrals are attached and may be removed for your convenience.

Name: _____ Teacher: _____

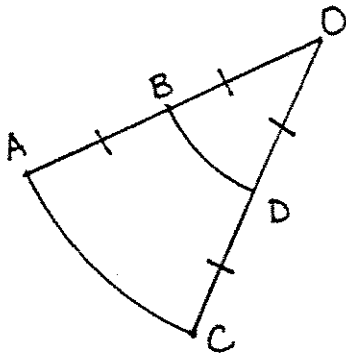
Question 1	Question 2	Question 3	Question 4	Question 5	Total
/12	/12	/13	/11	/12	/60

Question 1 (12 marks)

Marks

- a) Change 2.3 radians to degrees and minutes 1
- b) Find $\tan 3.2$ to 3 decimal places 1
- c) Find the exact value of
- i) $\sin \frac{3\pi}{4}$ 1
- ii) $\cos \left(\frac{-\pi}{6} \right)$ 1
- d) Solve $2 \sin x = -\sqrt{3}$ for $0 \leq x \leq 2\pi$ 2
- e) AC and BD are the arcs of two concentric circles with centre O.
 $AB = OB = CD = OD = 10\text{cm}$.

$$\angle AOC = \frac{\pi}{15}$$



Find i) the perimeter of Sector OAC 2
in terms of π

ii) the area of ABDC in terms 2
of π

- f) If $f(x) = 2 \cos 2x$ find $f' \left(\frac{\pi}{4} \right)$ 2

Question 2 (12 marks)

- a) Find
- i) $\frac{d}{dx}(\sin x^2)$ 1
- ii) $\frac{d}{dx}(\tan^2 x)$ 2
- b) The gradient function of a curve is given by $\frac{dy}{dx} = 6 \cos 2x - \sin x$.
If the curve passes through the origin, find the equation of the curve 3
- c) If $y = 3 \sin x - 4 \cos x$ show that $\frac{d^2y}{dx^2} + y = 0$ 3

d) Find $\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \cos 2x dx$ 3

Question 3 (13 marks)

a) Evaluate $\int_0^2 (1-2x)^3 dx$ 3

b) A curve is defined as follows

$$f(x) = \begin{cases} -1 & \text{for } -2 \leq x < 0 \\ x^2 - 1 & \text{for } 0 \leq x \leq 3 \end{cases}$$

i) Sketch the curve neatly showing a suitable scale on both axes 2

ii) State the co-ordinates of the point where the curve cuts the x axis. 1

iii) Find the area between the curve, the x axis and the lines $x = -2$ and $x = 3$. 4

c) Solve $2 \sin^2 x - \sin x = 0$ for $0 \leq x \leq 2\pi$ 3

Question 4 (11 marks)

a) i) Sketch $y = 4 \sin \frac{x}{2}$ in the domain $-2\pi \leq x \leq 2\pi$. State its period and amplitude 3

ii) Hence find the number of solutions to the equation $\sin \frac{x}{2} = \frac{1}{4}$ in the domain $-2\pi \leq x \leq 2\pi$ 1

b) i) The table below has been completed for $f(x) = \sqrt{\frac{x^2-1}{x}}$

x	1	1.25	1.5	1.75	2
f(x)	0	0.67	0.91	1.09	1.22

ii) By using Simpson's Rule and the five function values in the table find $\int_1^2 f(x) dx$ 2

- c) The area bounded by the curve $y = \sqrt{x^3}$, the x axis and the line $x = 3$, is rotated around the x axis. Find the volume of the solid formed (leave your answer in terms of π).

3

- d) Differentiate $y = x \sin 2x$

2

Question 5 (12 marks)

- a) i) Find the x values of the points of intersection of the line $y = 2x$ and the curve $3y = x^2$ 1
 ii) On the same axes sketch $y = 2x$ and $3y = x^2$, showing clearly the points of intersection. Label the line and the curve and shade the enclosed area. 2
 iii) Find the shaded area 2

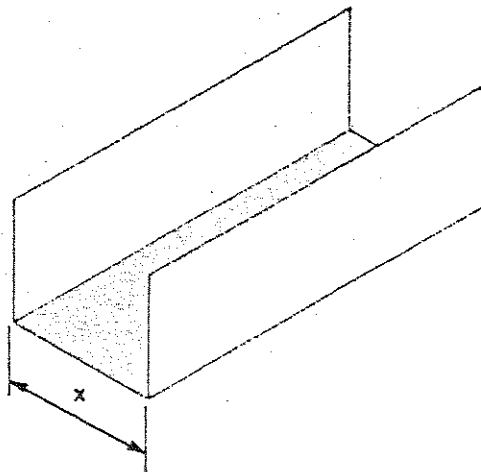
- b) A metal gutter open at the top and ends, is bent up from material 30cm wide to form a rectangular cross section.

- i) If the base of the gutter is x cm wide, find the height of the gutter. 1
 ii) Show that the area of the gutter cross section is given by

$$A = 15x - \frac{x^2}{2}.$$

1

- iii) Find the value of x for which A is a maximum. 3



- c) Find $\int \tan^2 \theta d\theta$

2

Question 1 (12 mks)

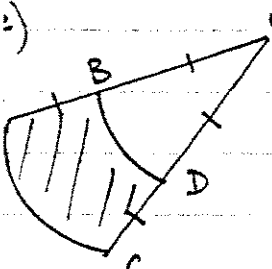
$$a) 2.3 \text{ radians} = \underline{131^\circ 47'} \quad (1)$$

$$b) \tan 3.2 = \underline{0.058} \quad (1)$$

$$c) i) \sin \frac{3\pi}{4} = \sin \left(\pi - \frac{\pi}{4} \right) \\ = \sin \frac{\pi}{4} \\ = \underline{\frac{1}{\sqrt{2}}} \quad (1)$$

$$ii) \cos \left(-\frac{\pi}{6} \right) = \cos \frac{\pi}{6} \\ = \underline{\frac{\sqrt{3}}{2}} \quad (1)$$

$$d) 2 \sin x = -\sqrt{3} \\ \sin x = -\frac{\sqrt{3}}{2} \quad \begin{array}{c|c} S & A \\ \hline \sqrt{T} & C \end{array} \\ \text{acute } x = \pi/3 \\ \therefore x = \underline{\frac{4\pi}{3}, \frac{5\pi}{3}} \quad (2)$$



$$e) i) P = 40 + 20 \times \frac{\pi}{15} \\ P = \underline{\left(40 + \frac{4\pi}{3} \right) \text{ cm}} \quad (2)$$

$$ii) A = \left(\frac{1}{2} \times 20^2 \times \frac{\pi}{15} \right) - \\ \left(\frac{1}{2} \times 10^2 \times \frac{\pi}{15} \right) \\ \therefore A = \frac{1}{2} \times \frac{\pi}{15} (20^2 - 10^2) \\ \underline{A = 10\pi \text{ cm}^2} \quad (2)$$

$$f) f(x) = 2 \cos 2x \quad (2) \\ f'(x) = -4 \sin 2x \\ \therefore f' \left(\frac{\pi}{4} \right) = -4 \sin \frac{\pi}{2} \\ = \underline{-4}$$

Question 2 (12 mks)

$$a) i) \frac{d}{dx} (\sin x^2) = \underline{2x \cos x^2} \quad (1)$$

$$ii) \frac{d}{dx} (\tan x)^2 = 2 \sec^2 x \cdot \tan x \quad (2)$$

$$b) \frac{dy}{dx} = 6 \cos 2x - \sin x$$

$$y = 3 \sin 2x + \cos x + c$$

sub (0,0)

$$0 = 3 \sin 0 + \cos 0 + c$$

$$\therefore c = -1$$

$$\underline{y = 3 \sin 2x + \cos x - 1} \quad (3)$$

$$c) y = 3 \sin x - 4 \cos x$$

$$\frac{dy}{dx} = 3 \cos x + 4 \sin x$$

$$\frac{d^2y}{dx^2} = -3 \sin x + 4 \cos x$$

$$\text{LHS} = \frac{d^2y}{dx^2} + y$$

$$= (-3 \sin x + 4 \cos x) + 3 \sin x - 4 \cos x$$

$$\underline{= \text{RHS}} \quad (3)$$

$$d) \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \cos 2x \, dx \quad (3)$$

$$= \left[\frac{1}{2} \sin 2x \right]_{\frac{\pi}{3}}^{\frac{2\pi}{3}}$$

$$= \frac{1}{2} \left[\sin \frac{4\pi}{3} - \sin \frac{2\pi}{3} \right]$$

$$= \frac{1}{2} \left[\sin \left(\pi + \frac{\pi}{3} \right) - \sin \left(\pi - \frac{\pi}{3} \right) \right]$$

$$= \frac{1}{2} \left[-\sin \frac{\pi}{3} - \sin \frac{\pi}{3} \right]$$

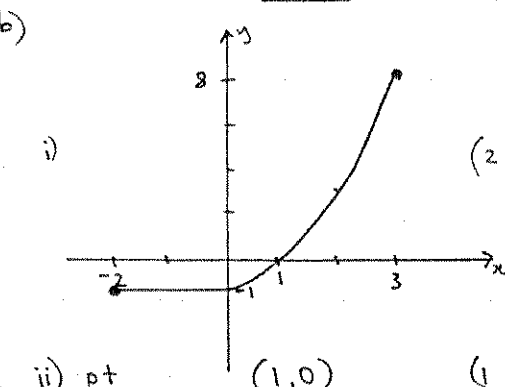
$$= \frac{1}{2} \left[-\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} \right] = \underline{-\frac{\sqrt{3}}{2}}$$

Question 3 (13 mks)

$$a) \int_0^2 (1-2x)^3 dx = \left[\frac{(1-2x)^4}{-8} \right]_0^2$$

$$= \frac{81}{-8} - \frac{1}{-8}$$

$$= -10 \quad (3)$$



iii)

$$A = 2 + \left| \int_0^1 (x^2 - 1) dx \right| + \int_1^3 (x^2 - 1) dx$$

$$= 2 + \left| \left[\frac{x^3}{3} - x \right]_0^1 \right| + \left[\frac{x^3}{3} - x \right]_1^3$$

$$= 2 + \left| \left(\frac{1}{3} - 1 \right) \right| + \left(6 - \frac{2}{3} \right)$$

$$= 2 + \frac{2}{3} + 6\frac{2}{3}$$

$$= 9\frac{1}{3} \text{ unit}^2 \quad (4)$$

2) $2 \sin^2 x - \sin x = 0$

$$\sin x (2 \sin x - 1) = 0$$

$$\sin x = 0 \quad 2 \sin x - 1 = 0$$

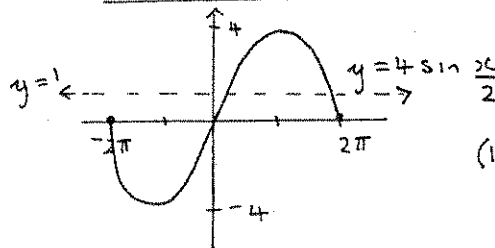
$$\sin x = \frac{1}{2}$$

$\therefore x = 0, \pi, 2\pi, \frac{\pi}{6}, \frac{5\pi}{6} \quad (3)$

Question 4 (11 mks)

a) i) amp = 4 (1)

period = 4π (1)



ii) $\sin \frac{x}{2} = \frac{1}{4}$

$$4 \sin \frac{x}{2} = 1$$

see $y=1$ above $\therefore$

2 solutions (1)

b) $\int_1^2 f(x) dx =$

$$= \frac{.25}{3} \left[0 + 1.22 + 4 \left(.67 + 1.09 \right) + 2x \cdot .91 \right]$$

$$= \underline{0.84} \quad (2)$$

c) $V_x = \pi \int_0^3 \left(\left(\frac{x^3}{3} \right)^2 \right) dx$

$$= \pi \int_0^3 x^3 dx$$

$$= \pi \left[\frac{x^4}{4} \right]_0^3$$

$$= \underline{\frac{81\pi}{4} \text{ unit}^3} \quad (3)$$

d) $u = x \quad v = \sin 2x$

$$u' = 1 \quad v' = 2 \cos 2x$$

$$\therefore \frac{d}{dx} (x \sin 2x) = \underline{\underline{\sin 2x + 2x \cos 2x}} \quad (2)$$

Question 5 (12 mks)

a) i) $y = 2x \quad 3y = x^2$

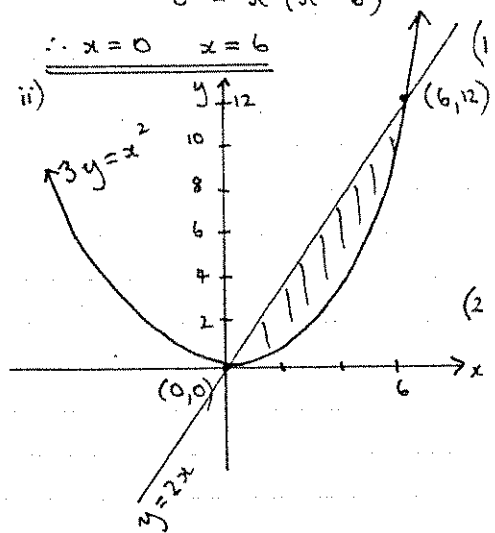
$$3(2x) = x^2$$

$$6x = x^2$$

$$0 = x^2 - 6x$$

$$0 = x(x-6)$$

$\therefore x = 0 \quad x = 6$ (1)



iii) $A = \int_0^6 \left(2x - \frac{x^2}{3} \right) dx$

$$= \left[\frac{2x^2}{2} - \frac{x^3}{9} \right]_0^6$$

$$= 36 - \frac{216}{9}$$

$$= \underline{12 \text{ unit}^2} \quad (2)$$

c) i) $2h + x = 30$

$$2h = 30 - x$$

$$h = \frac{30 - x}{2}$$

$\therefore \text{height} = \underline{\underline{\frac{30 - x}{2}}} \quad (1)$

ii) $A = x \left(\frac{30 - x}{2} \right)$

$$A = 15x - \frac{x^2}{2} \quad (1)$$

iii) $\frac{dA}{dx} = 15 - x$

$$\frac{d^2A}{dx^2} = -1$$

$\therefore$ st pt $\frac{dA}{dx} = 0 \quad 15 - x = 0 \quad (3)$

$$x = 15$$

when $x = 15 \quad \frac{d^2A}{dx^2} < 0 \therefore \text{max}$

c) $\int \tan^2 \theta d\theta$

since $1 + \tan^2 \theta = \sec^2 \theta$

$$\int \tan^2 \theta d\theta = \int (\sec^2 \theta - 1) d\theta$$

$$= \underline{\underline{\tan \theta - \theta + c}} \quad (2)$$